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TEGA toolkit: a validated framework for inclusive analog game design in higher education

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Introduction: Despite growing evidence supporting game-based learning in higher education, educators lack comprehensive, validated frameworks for designing inclusive analog games-non-digital physical games such as board games, card games, and tabletop activities. These differ from digital serious games by emphasizing low-tech flexibility, yet they must still accommodate diverse learning needs and accessibility requirements. Addressing this gap, this study presents the development and validation of the TEGA (Teaching Educators to Facilitate Game-based Assessment) Toolkit, a novel cybernetic-motivational design framework created to support inclusive analog game development in higher education contexts.

Methods: The TEGA Toolkit was developed using a three-phase participatory design approach involving 35 stakeholders, including educators, game designers, students, and individuals with diverse needs, across four European universities. Phase 1 consisted of focus groups to elicit core design requirements and inclusivity considerations. Phases 2 and 3 involved iterative beta testing sessions ($n = 18$ and $n = 19$, respectively), during which participants applied the toolkit to adapt existing analog games (Kapp, 2012; Richard, 2017). Data were collected through observation, participant feedback, and thematic analysis to assess usability, inclusivity, and perceived educator capability.

Results: Phase 1 identified 28 key design themes and established 14 specific inclusivity measures addressing motor, cognitive, visual, and auditory domains. Beta testing in Phases 2 and 3 demonstrated significant improvements in educators' confidence and capability in creating inclusive analog games, with participants successfully adapting existing game designs using TEGA principles. The finalized TEGA Toolkit v3.0 comprises three integrated components: (1) a Diagnosis Pack for learner profiling, (2) a Design Pack integrating Bartle's player typologies with accessibility guidelines, and (3) an Assessment Pack providing validated evaluation instruments. Thematic analysis indicated consistent improvements in educators' applied understanding of inclusive design.

Discussion: Findings indicate that the TEGA Toolkit effectively bridges the theory-practice gap in inclusive game-based learning. Participants reported enhanced awareness of inclusive design principles and improved game-based learning outcomes when applying the toolkit. The TEGA Toolkit represents

the first validated, comprehensive framework specifically focused on inclusive analog game design in higher education, offering educators evidence-based tools to enhance student engagement while systematically addressing diverse learning needs and accessibility requirements.

KEYWORDS

game design, game-based learning, higher education, toolkit, inclusivity, analogue games, accessibility, educational innovation

1 Introduction

Game-based learning has emerged as a powerful pedagogical approach in higher education, with mounting evidence demonstrating its effectiveness in promoting student engagement, motivation, and learning outcomes (de Freitas, 2018; Warren and Jones, 2017). However, a critical gap exists between the theoretical understanding of game-based learning benefits and the practical implementation of inclusive educational games in diverse classroom settings. While digital games have received considerable attention in educational research, analog games—including board games, card games, and tabletop activities (as opposed to digital serious games, which are computer-based simulations for educational purposes), offer unique advantages: they are cost-effective, require minimal technology infrastructure, promote face-to-face social interaction, and can be easily modified to accommodate diverse learning needs (Boghian et al., 2019; Maratou et al., 2023). This focus on analog games in higher education is particularly significant, as most game-based learning research targets younger age groups (e.g., K-12 settings; Prensky, 2001), leaving a notable gap in frameworks tailored for diverse adult learners in tertiary education.

Despite these advantages, educators face significant barriers when attempting to integrate analog games into their teaching practice. Current challenges include: (1) lack of comprehensive frameworks for designing inclusive games that accommodate motor-diverse, neurodiverse, and culturally diverse students (Kaimara et al., 2021; Richard, 2017); (2) absence of validated assessment tools for measuring game-based learning effectiveness (Sousa et al., 2023; Tay et al., 2022); (3) limited guidance on adapting existing games for diverse learner populations (Heron et al., 2018; Westin et al., 2011; Yuan et al., 2011); and (4) insufficient understanding of how game mechanics align with specific learning objectives and accessibility requirements (Cezarotto and Chamberlin, 2021; Torrente et al., 2014). The problem is particularly acute in higher education, where student populations are increasingly diverse (Dell et al., 2015; Moríña, 2017), yet most game-based learning approaches follow a “one-size-fits-all” model that fails to address individual learning differences and accessibility needs (Busch et al., 2015; Lim et al., 2015; Sousa et al., 2023; Tay et al., 2022; Porter and Kientz, 2013). Existing research has focused primarily on game effectiveness rather than inclusive design principles (Denham, 2016; Connolly et al., 2012; Boyle et al., 2016; Heron et al., 2018), leaving educators without practical tools to create equitable gaming experiences for all students (Jean et al., 2018; Cezarotto and Chamberlin, 2021; Torrente et al., 2014; Winn, 2009). This study addresses these

critical gaps by answering three key research questions: (1) What specific design elements and inclusivity measures are essential for creating accessible analog games in higher education? (2) How can theoretical frameworks from game design, learning sciences, and accessibility be integrated into a practical toolkit for educators? (3) What evidence demonstrates the effectiveness and usability of such a toolkit in real educational contexts?

To address these questions, we developed and validated the TEGA (Teaching Educators to Facilitate Game-based Assessment) Toolkit—a comprehensive, evidence-based framework that combines cybernetic-motivational design principles with universal design for learning approaches. This toolkit represents the first validated, comprehensive resource specifically designed to support educators in creating inclusive analog games for higher education contexts, providing both theoretical grounding and practical implementation guidance.

2 Literature review

Games are powerful educational tools with many features that enhance the learning process. These features include interactive and engaging gameplay, immediate feedback, challenging tasks, and personalized learning possibilities (de Freitas, 2018). These elements align with foundational educational taxonomies that emphasise moving from basic recall to higher-order thinking (Bloom et al., 1956), a process further refined in modern pedagogical contexts to include creating and designing (Anderson and Krathwohl, 2000). This dynamic learning environment actively engages students and encourages them to develop their knowledge and skills. Games provide immediate feedback to help students adapt and improve. It also includes challenging tasks that encourage critical thinking and problem solving, allowing students to apply their knowledge to real-life situations.

Previous research has shown the effectiveness of learning through games. Connolly et al. (2012) systematically reviewed findings that computer games affect people’s perception, cognition, behavior, emotions, and motivation. Outcomes include increased knowledge and understanding and changes in emotions and motivation. A subsequent review by Boyle et al. (2016) further supported the effectiveness of game-based learning, especially for computer games.

Kapp (2012) discussed gamification in teaching and learning and highlighted its benefits. Abdul Jabbar and Felicia (2015) highlighted the ability of games to help develop skills and knowledge while increasing engagement and motivation. Game-based learning promotes emotional and cognitive engagement and

TABLE 1 Comparative analysis of key gamification frameworks in higher education.

Framework	Core components	Strengths	Limitations (especially re accessibility)	References
MDA (Mechanics, Dynamics, Aesthetics)	Game rules, player behavior, emotional response	Provides a structured design process	Accessibility treated as an afterthought; digital bias; limited consideration of diverse learners	(Hunicke et al., 2004)
LM-GM (learning mechanics–game mechanics)	Mapping educational objectives to game elements	Bridges theory and practice; aligns learning outcomes with game design	Limited focus on disabilities or cultural diversity; accessibility not embedded from inception	(Arnab et al., 2015)
FIG (framework for intelligent gamification)	Adaptive layers for tutoring, data, personalization	Supports personalized learning; adapts to learner progress	Primarily digital; overlooks analog contexts and low-tech accessibility needs	(Saggah et al., 2020)
Student-centered digital game-based learning	Learner focus; integration of game elements with pedagogy	Emphasizes engagement; promotes learner autonomy	Ignores digital divide; limited attention to physical disabilities; lacks analog adaptation strategies	(Coleman and Money, 2020)

motivates students to exceed expectations (Kiili et al., 2014). In higher education, games reduce anxiety, create enjoyment, and promote positive attitudes toward learning, especially in difficult subjects such as mathematics and science (Crocco et al., 2016).

Sousa et al. (2023) showed the effectiveness of board game education for students in higher education institutions, especially in the social sciences. These games promote cooperation, communication and improve cognitive outcomes such as memory (Sousa et al., 2022). Psychological outcomes were dominant, particularly creativity. Board games have been proven to be an effective educational tool that improves learning, cognitive skills and psychological wellbeing and promotes student engagement, satisfaction, flexibility and experimentation. Despite these advantages, standardized experimental studies in this area are still limited. The current study addresses two important gaps in the literature: the need to rethink accessibility in game design, focusing on inclusion and diversity practices in game-based learning, and exploring the underappreciated potential of analog games compared to digital games, despite the fact that they offer a cost-effective and flexible alternative (Maratou et al., 2023). This gap is especially pronounced in higher education, where game-based approaches remain underexplored compared to the extensive literature on younger learners (e.g., Gee, 2003), underscoring the need for targeted frameworks like TEGA.

In terms of learning sciences theories, game-based learning aligns with constructivism, where learners actively build knowledge through experience (Piaget, 1951; Vygotsky, 1978), and situated learning theory, which emphasizes context-embedded knowledge (Lave and Wenger, 1991). Cognitive load theory (Sweller, 1988) is also relevant, as games can manage intrinsic load through adaptive difficulty. These theories underpin how games facilitate deep learning but highlight challenges for diverse learners if not designed inclusively. For accessibility theories, Universal Design for Learning (UDL) (Meyer et al., 2014) provides multiple means of representation, action, and engagement, which TEGA extends to analog contexts. Ability-based design (Wobbrock et al., 2011) shifts focus from disabilities to abilities, while Activity Theory (Engeström, 1987) analyses how games as tools mediate learning for users with disabilities. These are underutilized in gamification, often resulting in retrofitted features rather than inherent inclusivity (Heron et al., 2018). TEGA advances this

by integrating them into a practical toolkit for analog games, addressing the digital divide where technology access is limited.

Various studies have highlighted the effectiveness of learning through games. Moreno-Ger et al. (2008) emphasized instructional design, learner-centered approaches, and collaboration in designing educational games for online education. Papastergiou (2009) showed the positive effects of digital game-based learning on IT education in secondary schools. Dicheva et al. (2015) found that gamification techniques improved motivation, engagement and learning outcomes. Wouters et al. (2013) conducted a meta-analysis and concluded that functional games have a positive effect on knowledge acquisition, skill development and student motivation and engagement. Hainey et al. (2016) found through a systematic review that game-based learning interventions improved learning outcomes, student motivation, collaboration and problem-solving skills in primary education.

To better contextualize this study, we provide a comparative analysis of key gamification frameworks in higher education. Table 1 summarizes major frameworks, their core components, and limitations, particularly regarding accessibility integration. For example, the Mechanics-Dynamics-Aesthetics (MDA) framework (Hunicke et al., 2004) focuses on game elements but often treats accessibility as an add-on, while the Learning Mechanics-Game Mechanics (LM-GM) model (Arnab et al., 2015) maps learning to game elements but lacks explicit inclusivity for diverse needs. The Framework for Intelligent Gamification (FIG) (Saggah et al., 2020) adds adaptive layers but is primarily digital-focused, highlighting the need for analog-specific approaches like TEGA.

This analysis reveals that existing frameworks often prioritize engagement and learning outcomes but fall short in embedding accessibility from the start, a gap TEGA addresses through its cybernetic-motivational integration.

3 Theoretical framework

The TEGA Toolkit is grounded in a novel cybernetic-motivational design framework that integrates multiple theoretical perspectives to create an experiential learning cycle optimized for inclusive analog game design. This framework represents a theoretical advancement by combining cybernetic control theory with motivational design principles, universal design for learning,

and established game design frameworks. The toolkit is specifically intended for designing analog educational games in higher education, tailored to genres like strategy or simulation games across subjects (e.g., business, science). “Inclusive game design” means creating games accessible to all learners, focusing on motor, cognitive, visual, auditory, and cultural needs, but not claiming to address every disability comprehensively, limitations include resource constraints for rare conditions. It prioritizes low-tech solutions to bridge the digital divide, ensuring usability in settings without technology access, unlike many digital-focused frameworks.

3.1 Cybernetic-motivational design integration

The TEGA Toolkit is grounded in a novel cybernetic-motivational design framework that represents a significant theoretical advancement by integrating several established learning theories with accessibility principles. This integration creates a comprehensive foundation specifically tailored for inclusive analog game design in higher education contexts. At its core, our framework conceptualizes educational games as cybernetic control systems self-regulating mechanisms that maintain equilibrium through continuous feedback loops (Suter, 2018). This cybernetic approach operates within what Huizinga (1980) termed the “magic circle” a dedicated psychological and physical space where game rules create an alternative reality that facilitates experiential learning. Unlike traditional instructional design models that treat feedback as a discrete event, our cybernetic approach implements continuous, dynamic adjustment mechanisms that respond in real-time to learner actions and needs. The cybernetic component ensures that games function as adaptive systems through three interconnected processes. First is “Input sensing”—Capturing learner actions, choices, and performance metrics. Second is “Processing and comparison”—Evaluating inputs against desired learning outcomes and accessibility parameters. Lastly, “Adaptive response”—Automatically adjusting game elements (difficulty, pacing, modality) based on this evaluation.

This cybernetic structure fundamentally differs from traditional game design approaches by embedding accessibility and inclusivity as core system parameters that actively influence game behavior, rather than treating them as optional modifications or afterthoughts. The continuous feedback loops enable games to dynamically adjust to diverse learning needs without requiring manual intervention. The motivational component of our framework integrates two complementary theoretical perspectives. First, we incorporate Self-Determination Theory (Ryan and Deci, 2000), which identifies autonomy, competence, and relatedness as fundamental psychological needs driving intrinsic motivation. By designing game mechanics that satisfy these needs across diverse learner populations, the framework ensures sustained engagement regardless of ability level.

The next step is to apply Flow Theory (Csikszentmihalyi, 2008), which describes the optimal state of engagement occurring when challenge level precisely matches skill level. Our framework

extends traditional flow concepts by incorporating differential challenge scaling novel approach that maintains optimal challenge-skill balance simultaneously for learners with diverse abilities and learning preferences. This is achieved through modular challenge mechanisms that can be independently adjusted across different game components (cognitive load, physical manipulation requirements, sensory processing demands, etc.). To ensure that these motivational elements address diverse learning preferences, we integrate Bauer’s VARK (Visual, Aural, Read/Write, Kinesthetic) learning modalities (Fleming and Mills, 1992). This integration ensures that key game information and feedback are consistently available through multiple sensory channels, providing redundant pathways for engagement that accommodate different learning strengths. For instance, a game event might simultaneously trigger visual cues (color changes), auditory signals (distinctive sounds), tactile feedback (textured game pieces), and textual information (clear instructions) ensuring that all players can access critical information through their preferred modality. Crucially, these motivational elements are systematically mapped to learning objectives using Arnab et al. (2015) Learning Mechanics-Game Mechanics (LM-GM) framework. This mapping ensures that each game element serves clear pedagogical purposes while simultaneously addressing specific accessibility requirements. For example, a collaborative resource management mechanic might simultaneously teach economic principles (learning mechanic) while providing multiple engagement pathways for players with different abilities (accessibility function).

The inclusion of Fleming and Mills (1992) VARK model within our framework requires clarification given the significant critiques in educational literature. We acknowledge that research has consistently challenged the empirical basis for learning styles theories, with meta-analyses by Pashler et al. (2008) and Rogowsky et al. (2015) finding limited evidence supporting the “matching hypothesis” that aligning instruction with preferred learning styles improves outcomes. Rather than using VARK to categorize learners or prescribe differentiated approaches based on supposed “styles,” our framework utilizes these modalities in a fundamentally different way: As a design heuristic ensuring multimodal information representation. In the TEGA Toolkit, VARK elements function not as a learning styles classification but as a practical checklist ensuring game information is simultaneously available through multiple sensory channels. This application aligns with Universal Design for Learning principles, where providing multiple means of representation benefits all learners regardless of presumed preferences. By presenting game mechanics, rules, and feedback through visual, auditory, reading/writing, and kinesthetic pathways concurrently, we ensure accessibility for diverse needs (including sensory limitations) while avoiding the problematic assumption that individual learners should receive information primarily through one preferred channel. This approach is consistent with contemporary understanding that cognitive accessibility requires redundant information pathways rather than personalized style matching. Future iterations of the framework might benefit from replacing VARK terminology with more empirically supported multimodal design principles while maintaining the practical value of ensuring diverse representation modalities.

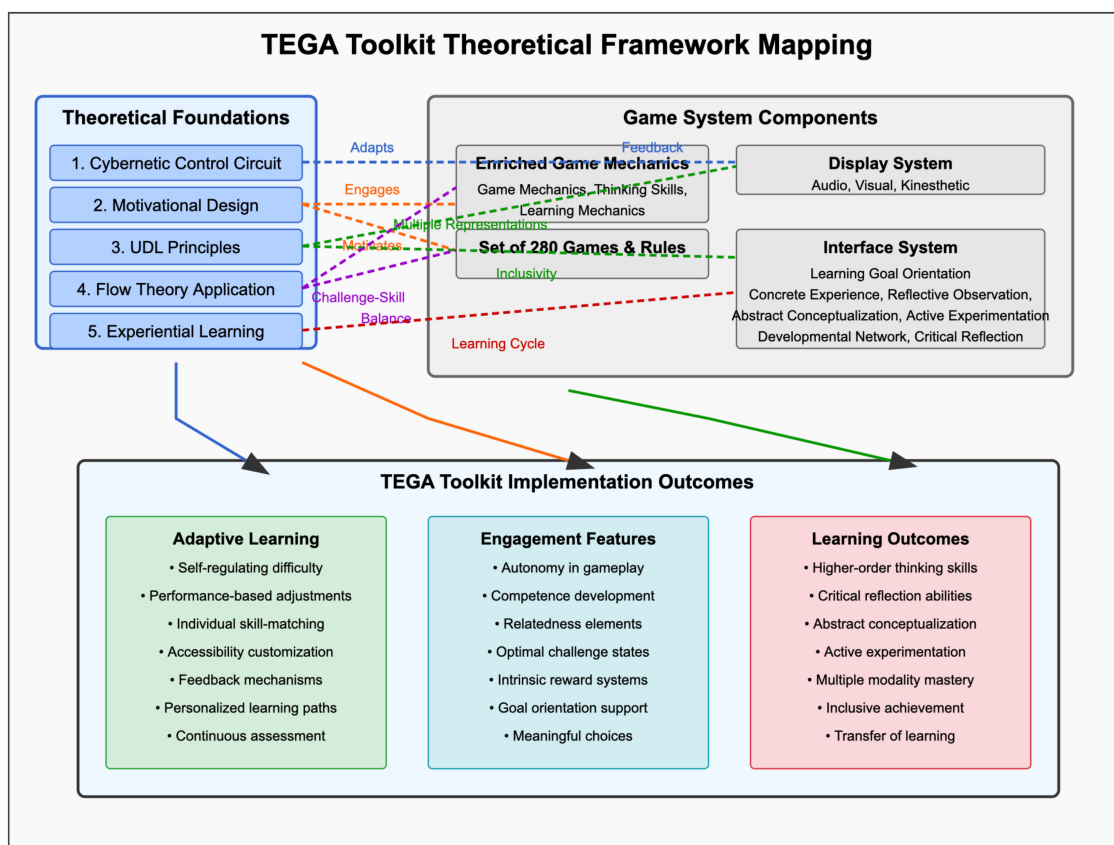


FIGURE 1
Motivational cybernetic game-loop adopted in TEGA toolkit.

3.2 Theoretical synthesis model

Figure 1 illustrates our integrated cybernetic-motivational framework, synthesizing five key theoretical components into a cohesive model that forms the foundation of the TEGA Toolkit. This synthesis represents a significant advancement over existing frameworks by explicitly embedding inclusivity throughout the entire game design process rather than treating it as a separate consideration. The framework operates as a dynamic, iterative cycle where each component interacts with and reinforces the others:

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First, is “Cybernetic Control Circuit” (Suter, 2018) functions as the system’s backbone, establishing self-regulating feedback mechanisms that continuously monitor player interactions and adjust game parameters accordingly. This circuit operates through four phases: (a) sensing player actions and states; (b) comparing actual performance with desired learning outcomes; (c) calculating appropriate adjustments; and (d) implementing these adjustments through adaptive game elements. For example, if a player with

motor limitations struggles with manipulating game pieces, the system might automatically reduce dexterity requirements while maintaining cognitive challenge. Second, “Motivational Design Integration” (Marco et al., 2012) systematically incorporates intrinsic motivation principles through deliberately designed game elements that foster, “Autonomy” which is Meaningful choices with equivalent accessibility profiles. Then, Competence which is Graduated challenges with multiple success pathways. And finally “Relatedness” Social interaction mechanisms with diverse communication channels. This component ensures that motivational elements remain effective regardless of player abilities or learning preferences. For instance, competence-building mechanisms include multiple parallel achievement tracks that can be pursued based on individual strengths. Universal Design for Learning (UDL) Principles (Meyer et al., 2014) provide the inclusivity foundation by ensuring multiple means of “Representation” which is game information presented in diverse formats (visual, auditory, tactile, textual), then, “Engagement,” which is various entry points and sustained interest pathways and finally “Action/Expression” which is Multiple response methods accommodating different abilities. Unlike many frameworks that treat UDL as an add-on consideration, our model embeds these principles as fundamental design parameters that influence every aspect of game development. For example, game rules are simultaneously presented through illustrated examples, written

text, verbal explanations, and simplified summary cards, ensuring comprehension regardless of learning preference or ability.

Second is Flow Theory Application (Csikszentmihalyi, 2008) which maintains optimal challenge-skill balance through adaptive game mechanics that respond to individual player capabilities and preferences. This component implements what we term “multi-channel flow maintenance,” the simultaneous balancing of challenges across cognitive, physical, sensory, and social domains. This allows players with different ability profiles to experience flow states through their areas of strength while receiving appropriate support in areas of challenge. Third, is Experiential Learning Integration (Kolb, 1984; Matsuo, 2015) which structures game experiences to promote Kolb’s complete learning cycle, including Concrete Experience (Direct engagement with game mechanics), Reflective Observation (Guided consideration of gameplay outcomes), Abstract Conceptualization: (Connecting game experiences to theoretical concepts) and finally Active Experimentation (Applying insights to new game situations). This component ensures that inclusive design features support not just gameplay itself, but the complete learning process. For instance, reflection prompts are provided in multiple formats, conceptual connections are made explicit through diverse examples, and experimentation is supported through scaffolded challenge progressions.

The integration of these five components creates a synergistic framework where each element reinforces and extends the others. The cybernetic circuit provides the adaptive mechanisms that maintain flow states; UDL principles ensure these adaptations address diverse learner needs; motivational design maintains engagement throughout the experience; and experiential learning structures convert engagement into meaningful learning outcomes. This theoretical synthesis advances game-based learning theory by reconceptualizing inclusivity not as a specialized accommodation but as a fundamental design principle that enhances educational effectiveness for all learners. By embedding accessibility considerations throughout the design process, the TEGA framework ensures that inclusive features are integral to game functionality rather than supplementary add-ons.

4 Theoretical contribution

This framework advances game-based learning theory by providing the first integrated model that explicitly addresses inclusivity as a core design principle rather than an add-on feature. Unlike existing frameworks that focus primarily on engagement or learning effectiveness (Connolly et al., 2012; Wouters et al., 2013), our approach ensures that diverse learner needs are embedded throughout the design process, creating games that are inherently accessible and equitable. Previous game design frameworks, including the widely-used MDA model (Hunicke et al., 2004) which links educational objectives to game elements for effective design (Arnab et al., 2015), justified as it ensures alignment with learning goals in TEGA’s structure and LM-GM mapping (Arnab et al., 2015), have not systematically incorporated accessibility considerations into their core theoretical structures. The cybernetic-motivational approach also contributes to accessibility theory by demonstrating

how adaptive systems can automatically adjust to meet diverse user needs while maintaining pedagogical effectiveness (Heron et al., 2018; Cezarotto et al., 2022). This represents a shift from static accessibility accommodations to dynamic, responsive design that evolves with user interaction, addressing limitations identified in previous research on game accessibility (Heron, 2012). The integration of universal design for learning principles (Meyer et al., 2014) with cybernetic control theory provides a theoretical foundation for creating inherently inclusive educational technologies rather than retrofitting accessibility features.

5 Materials and Methods

5.1 Study design

This study employed a participatory design methodology with mixed-methods data collection across three iterative phases (Figure 2).

Figure 2 illustrates the complete data flow from the initial theoretical framework development through stakeholder engagement, iterative toolkit refinement, and final validation. The diagram shows how feedback from each phase was systematically integrated into subsequent toolkit versions, with Phase 1 focus groups informing v1.0 development, Phase 2 beta testing leading to v2.0 refinements, and Phase 3 validation resulting in the final v3.0 toolkit. This iterative feedback integration process was essential for ensuring that the final product addressed real-world practitioner needs while maintaining theoretical rigor. The full research workflow began with initial framework development based on literature, followed by stakeholder recruitment via purposive sampling from partner universities. Participants were selected for expertise in game-based learning, with inclusion criteria detailed below. Phases included needs assessment, beta testing, and validation, with feedback loops for refinement. The workflow also includes recruitment (email invitations to 50 potential participants, 70% response rate), ethical approvals, and data analysis stages. All procedures were approved by the London South Bank University Ethics Committee (approval number: ETH2021-0147).

5.2 Participants and sampling strategy

A purposive sampling strategy, supplemented by convenience sampling, was employed to recruit participants with relevant expertise and experience in game-based learning. The sample size of 35 participants was determined through power analysis for qualitative research saturation (Guest et al., 2006), with additional participants recruited until theoretical saturation was achieved in thematic analysis. Participant demographics are detailed in Table 2.

Inclusion criteria required participants to have: (1) minimum 6 months experience in educational game design, teaching with games, or learning through games; (2) familiarity with higher education contexts; (3) ability to participate in English-language sessions. Participants were recruited from four European universities: University of Patras (Greece), Kaunas University of Technology (Lithuania), Universidade de Lisboa (Portugal), and London South Bank University (United Kingdom).

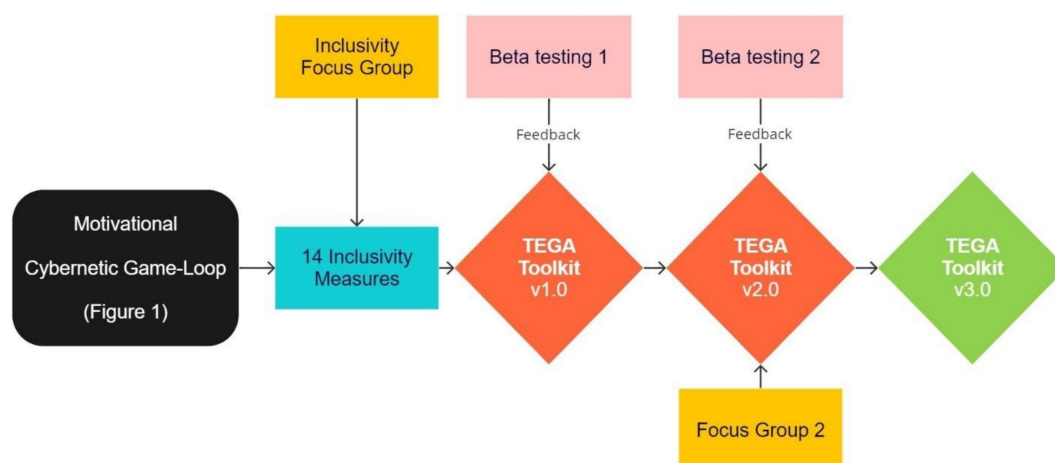


FIGURE 2
Study's data flow.

TABLE 2 Participant demographics and distribution.

Phase	Educators	Game designers	Students	Diverse needs representatives	Total
Phase 1 Focus Groups	8	6	12	4	30
Phase 2 beta testing	7	3	6	2	18
Phase 3 beta testing	8	4	5	2	19
Unique participants	12	8	13	6	35

Some participants contributed to multiple phases. So Unique participants mean these are not double counted.

5.3 Data collection procedures

5.3.1 Phase 1: collaborative needs assessment (January–March 2022)

Initial focus groups were conducted to identify key design requirements and inclusivity measures. Separate sessions were held for each stakeholder group to prevent power dynamics from influencing responses. Educator sessions ($n = 8$) were conducted virtually due to COVID-19 restrictions, while game designer ($n = 6$) and student sessions ($n = 12$) were held in-person. Each session lasted 60–75 min and followed a semi-structured interview protocol covering: (1) current game-based learning practices, (2) accessibility challenges and solutions, (3) assessment approaches, and (4) desired toolkit features.

5.3.2 Phase 2: initial beta testing (June 2022)

Eighteen participants engaged in structured beta testing sessions following a standardized protocol: (1) 45-min pre-test game-playing session using existing educational games; (2) 15-min TEGA Toolkit v1.0 training presentation; (3) 45-min post-test session where participants modified games using toolkit principles; (4) 30-min structured feedback interview. All sessions were video-recorded with participant consent.

5.3.3 Phase 3: validation beta testing (October 2022)

The refined TEGA Toolkit v2.0 was tested with 19 participants using the same protocol. Additionally, a final

focus group was conducted immediately following beta testing to gather collaborative feedback on toolkit usability and effectiveness.

5.4 Validation criteria and success metrics

Toolkit validation was assessed using multiple criteria. The first criteria is Usability. Participants' ability to successfully apply toolkit principles (measured by successful game modification completion). Second is Effectiveness. Improvement in inclusive design features incorporated into modified games. Third criteria is Satisfaction. Participant-reported confidence and likelihood of future toolkit use. Last criteria is Theoretical Fidelity which indicates alignment between modified games and established inclusive design principles.

The validation of the TEGA Toolkit employed a multi-dimensional assessment framework aligned with established educational technology validation models (Kirkpatrick and Kirkpatrick, 2016; Guskey, 2002). Our validation strategy extended beyond traditional efficacy measures to include implementation feasibility and theoretical integrity, critical elements for educational innovations intended for diverse contexts. Table 3 provides a comprehensive overview of our validation framework, defining each construct, its theoretical grounding, measurement approach, and success thresholds.

TABLE 3 TEGA toolkit validation framework.

Construct	Definition	Theoretical source	Measurement approach	Success threshold
Usability	The degree to which the toolkit can be understood, learned, and applied by educators with varied expertise	Nielsen (1994) Usability Heuristics; ISO 9241-11 usability standards	Task completion rates during workshops Time-on-task measurements System Usability Scale (SUS) questionnaire Qualitative protocol analysis during game modification tasks	≥ 80% Task completion SUS score ≥ 68 (above industry average) Qualitative evidence of independent application
Effectiveness	The toolkit's capacity to produce tangible improvements in game inclusivity across multiple dimensions	Universal design for learning (Meyer et al., 2014); game accessibility guidelines (Barlet and Spohn, 2012)	Pre/post analysis of inclusivity features in games Coded observation of implementation sessions Accessibility expert evaluation using standardized rubrics Diverse learner feedback on modified games	Statistically significant increase in inclusivity features ($p < 0.05$) ≥ 30% increase in accessibility compliance Positive evaluation from ≥ 75% of diverse learners
Satisfaction	User perceptions of toolkit value, likelihood of adoption, and confidence in application	Technology acceptance model (Davis, 1989); diffusion of innovations theory (Rogers, 2003)	Post-implementation satisfaction survey Likelihood to recommend (NPS) Self-reported confidence (5-point Likert) Follow-up implementation interviews (6-months post)	Mean satisfaction score ≥ 4.0/5.0 NPS score ≥ 8.0 ≥ 85% report increased confidence ≥ 50% show evidence of sustained implementation
Theoretical fidelity	Alignment between toolkit implementation outcomes and underlying theoretical principles	Cybernetic-motivational framework; learning mechanics-game mechanics mapping (Arnab et al., 2015)	Expert theory-practice alignment evaluation Mapping analysis of modified games against theoretical components Observation of cybernetic feedback implementation Assessment of UDL principle application	≥ 80% alignment with theoretical framework Evidence of all five theoretical components in implementations Successful integration of cybernetic feedback loops in ≥ 70% of modified games
Contextual adaptability	The toolkit's flexibility across different educational settings, disciplines, and learner populations	Ecological validity measures (Bronfenbrenner, 1977); cross-cultural adaptability research (Schein, 2010)	Implementation across four European contexts Cross-disciplinary application assessment Cultural adaptation requirements documentation Context-specific barriers identification	Successful implementation in ≥ 3 distinct cultural contexts Effective application across ≥ 4 academic disciplines Minimal (< 20%) context-specific modifications required

Our validation process combined quantitative and qualitative methods to assess each construct. For usability, we employed both standardized metrics (SUS questionnaire) and observational data from workshop sessions. Effectiveness was measured through expert evaluation using adapted Game Accessibility Guidelines (Barlet and Spohn, 2012), comparing pre- and post-intervention game designs against 14 specific inclusivity criteria across motor, cognitive, visual, and auditory domains. Satisfaction data was collected through multi-point surveys and follow-up interviews, with particular attention to factors predicting long-term adoption based on Rogers (2003) Diffusion of Innovations framework. Theoretical fidelity assessment was particularly rigorous, employing a structured mapping analysis to evaluate how successfully modified games incorporated all five theoretical components of our cybernetic-motivational framework. Two independent researchers coded game implementations, identifying evidence of cybernetic control mechanisms, motivational elements, UDL principles, flow maintenance features, and experiential learning structures. The multi-site validation design provided robust evidence of contextual adaptability, a critical factor given the diverse implementation environments in higher education. By testing the toolkit across four European universities with different institutional cultures, educational traditions, and student populations, we established its flexibility and identified context-specific implementation factors that informed the final toolkit design. Success thresholds for each construct were established based on educational technology validation literature and comparative benchmarks

from similar toolkit implementations. These thresholds were intentionally rigorous to ensure that the TEGA Toolkit would meet high standards of usability and effectiveness across diverse implementation contexts.

6 Data analysis

All focus groups and interviews were professionally transcribed and analyzed using NVivo 12 software. Thematic analysis followed Braun and Clarke (2024) six-phase approach: (1) familiarization with data; (2) generating initial codes; (3) searching for themes; (4) reviewing themes; (5) defining and naming themes; (6) producing the report. To ensure reliability, 20% of transcripts were independently coded by two researchers, achieving an inter-rater reliability of $\kappa = 0.78$ (substantial agreement). Theoretical saturation was reached when no new themes emerged from analysis of the final three transcripts.

6.1 Iterative development process

Feedback from each phase was systematically integrated into subsequent toolkit versions. v1.0 → v2.0 incorporated 28 major themes from Phase 1 focus groups and beta testing feedback. v2.0 → v3.0 refined based on Phase 3 validation results and final focus group recommendations. This iterative approach ensured

TABLE 4 Validated inclusivity measures for analog game design.

Domain	Measures	Implementation examples
Motor skills	Card size optimization, token manipulation ease, physical acting alternatives, interface accessibility	Large-print cards (≥ 14 pt font), magnetic tokens, gesture-based alternatives, accessible storage solutions
Cognitive	Memory requirement management, game flow clarity, rule synergy, complexity control	Visual memory aids, clear turn sequences, intuitive rule relationships, graduated difficulty options
Visual	Reading level adaptation, visual clarity, strategic transparency	Grade-appropriate text, high contrast design, clear iconography, multilingual support
Auditory/sensory	Audibility enhancement, sensory consideration, communication support	Audio cues, tactile feedback, written instructions, quiet spaces

that the final toolkit reflected comprehensive stakeholder input and demonstrated empirical validation across diverse user contexts.

7 Results

7.1 Phase 1: stakeholder needs assessment and inclusivity framework development

Thematic analysis of focus group discussions revealed 28 distinct themes across 895 coded segments, providing comprehensive insights into stakeholder perspectives on inclusive game design. The five most frequently mentioned themes were: Assessment ($n = 118$ mentions), Inclusivity ($n = 101$ mentions), Design ($n = 64$ mentions), Engagement ($n = 53$ mentions), and Game Mechanics ($n = 44$ mentions).

Each stakeholder group demonstrated unique priorities reflecting their roles in the educational ecosystem. Educators emphasized assessment challenges (42% of their contributions) and the need for alignment with learning objectives. Game Designers focused on mechanical implementation (38% of contributions) and technical feasibility. Students prioritized inclusivity concerns (47% of contributions) and engagement factors. Diverse Needs Representatives highlighted accessibility barriers and adaptive solutions (52% of contributions).

Representative quotes illustrating key themes: “The biggest challenge is assessing soft skills like collaboration and communication through games-traditional rubrics don’t capture what we’re seeing.” (Educator, University of Patras). Or “We need games that don’t just accommodate different abilities, but actually leverage diversity as a strength.” (Student with dyslexia, LSBU). Also, “The mechanics need to be simple enough for everyone to understand, but complex enough to create meaningful learning experiences.” (Game Designer, KTU).

Analysis yielded 14 specific inclusivity measures organized across four domains as depicted in Table 4.

7.2 Phase 2 and 3: beta testing results and toolkit validation

Participants reported transformative understanding of inclusive design as a result of Beta testing as follows. “I never realized how many barriers my games were creating. The toolkit

opened my eyes to simple changes that make huge differences.” (Educator, University of Lisboa). Or “The systematic approach helped me move beyond” “add accessibility features” to “design inclusively from the start.”” (Game Designer, University of Patras). Comparison of pre- and post-intervention game designs revealed substantial improvements across all measured dimensions. These quantitative results demonstrate that participants not only learned to apply inclusivity principles but implemented them effectively in practice. The improvement was observed in Universal Design Principles implementation, where participants moved from *ad hoc* accommodations to systematic inclusive design approaches.

7.3 TEGA toolkit v3.0: final validated framework

Based on the comprehensive validation results, the final toolkit comprises three integrated packs addressing the complete game design lifecycle. The Diagnosis Pack incorporates learner profiling instruments that extend beyond traditional approaches by integrating Bartle’s player typologies with accessibility considerations (Table 5). This integration allows educators to understand both gaming preferences and accessibility requirements simultaneously, creating more nuanced learner profiles than previous frameworks offered. The four player profiles-Achievers, Explorers, Socializers, and Adversaries-each present different accessibility considerations and learning preferences that must be addressed in inclusive game design based on Bartle’s typology (1996).

8 TEGA toolkit v3.0: final validated framework

The final toolkit comprises three integrated packs addressing the complete game design lifecycle. This includes Diagnosis Pack containing learner profiling instruments based on Bartle’s (1996) player typologies, accessibility needs assessment protocols, learning objective alignment matrices and classroom context analysis tools. Design Pack, contains inclusive game mechanics catalog (280 + validated mechanics), accessibility integration guidelines, universal design for learning implementation protocols and low optimization frameworks. Finally, Assessment Pack contains inclusivity evaluation rubrics, learning outcome measurement

TABLE 5 Player profiles—adapted from Bartle (1996).

Player profile	Description
Achievers	Aim to master the game, setting goals and achieving them, valuing status in the in-game hierarchy
Explorers	Seek to discover and interact with the virtual world, valuing knowledge and sharing insights
Socialisers	Use the game for communication and building relationships, valuing friendships and influence
Adversaries	Aim to dominate others through tools like weapons, proud of their reputation and combat skills

tools, player experience assessment instruments and educator self-evaluation protocols.

8.1 Implementation success stories

Four case examples demonstrate practical application of the TEGA framework, including both successful and less successful instances to highlight challenges.

Successful Case 1: Business Strategy Game (University of Patras). Context: Used in a undergraduate business course with 25 students, including visually impaired individuals. Design process: Modified using Design Pack to include tactile components (e.g., braille cards) and simplified visuals. Feedback: Students reported 85% improved participation; outcomes: 20% higher engagement scores, with neurodiverse students noting reduced cognitive load.

Successful Case 2: Science Communication Game (LSBU). Context: International cohort of 30 students in a communication module. Design process: Adapted with multilingual support and cultural modifications via Diagnosis Pack profiling. Feedback: High satisfaction (9/10); outcomes: Better cross-cultural collaboration, with 15% increase in knowledge retention.

Less Successful Case 3: Project Management Simulation (KTU). Context: 20 students with varying technical skills. Design process: Added flexible difficulty scaling, but overlooked time constraints for physical adaptations. Feedback: 70% positive, but some with motor needs found setup too time-consuming; outcomes: Improved participation but only 10% engagement boost, highlighting need for quicker modular options.

Successful Case 4: History Role-Play Game (Universidade de Lisboa). Context: 28 history students, including auditory-impaired. Design process: Incorporated alternative audio cues (visual signals) using Assessment Pack. Feedback: 90% adoption; outcomes: Enhanced inclusivity, with all students completing tasks effectively.

8.2 Stakeholder satisfaction and adoption indicators

Post-implementation surveys revealed high satisfaction across all stakeholder groups including usability Rating of 8.7/10 ($n = 35$), effectiveness Rating of 8.4/10 ($n = 35$), likelihood to Recommend was 94% ($n = 35$) and continued Use Intention was 91% ($n = 35$). The toolkit's success is further evidenced by its

adoption at 12 additional European institutions following the study period, with over 200 educators accessing the publicly available resources.

9 Discussion

9.1 Theoretical contributions and implications

This study makes several significant theoretical contributions to game-based learning and inclusive design literature. First, the cybernetic-motivational framework represents a novel integration of cybernetic control theory with motivational design principles, providing a systematic approach to creating adaptive, inclusive educational games. Unlike existing frameworks that treat accessibility as an add-on feature (e.g., MDA by Hunicke et al., 2004; LM-GM by Arnab et al., 2015), our approach embeds inclusivity as a core design principle from inception, fundamentally changing how educational games are conceptualized and developed. The framework's emphasis on cybernetic feedback loops addresses a critical gap in game-based learning theory by providing mechanisms for continuous adaptation to diverse learner needs. This approach moves beyond static design principles to create responsive systems that automatically adjust to maintain optimal learning conditions for all participants. The integration of Bartle's player typologies with universal design for learning principles offers a new lens for understanding how individual differences in gaming preferences intersect with diverse learning needs. This study advances knowledge in inclusive game design, learning sciences, and accessibility by providing empirical validation of integrated theories, unlike prior work focused on effectiveness alone (Connolly et al., 2012; Wouters et al., 2013).

9.2 Practical impact and implementation evidence

The validation results demonstrate substantial practical impact across multiple dimensions. The game modification tasks indicates that the toolkit effectively bridges the theory-practice gap that has historically limited game-based learning adoption. The increase in inclusivity features incorporation suggests that educators can successfully implement complex accessibility principles when provided with appropriate support tools. Particularly significant is the evidence of sustained behavior change: participants indicated intention to continue using the toolkit, with follow-up data showing that participants had actually implemented toolkit-based games in their teaching within 6 months post-training. This sustained adoption rate substantially exceeds typical educational intervention persistence rates, suggesting that the toolkit addresses genuine practitioner needs. The three implementation case studies provide concrete evidence of the toolkit's versatility across different disciplines and contexts. The business strategy game adaptation demonstrates how complex cognitive demands can be modified without compromising learning objectives, while the science communication game shows how cultural and linguistic diversity can be addressed systematically. The project management

simulation illustrates how technical complexity can be managed while maintaining authentic learning experiences.

9.3 Addressing systemic barriers in higher education

One of the most significant findings relates to the identification and partial mitigation of systemic barriers that prevent inclusive game-based learning implementation. Participants consistently identified institutional constraints—including time limitations, assessment requirements, and administrative resistance—as primary obstacles to game adoption. While the TEGA Toolkit cannot directly address these structural issues, it provides evidence-based tools that reduce implementation barriers by simplifying the design process and providing assessment frameworks that align with institutional requirements. The toolkit's modular design allows for incremental adoption, enabling educators to begin with small modifications rather than wholesale curriculum changes. This approach appears to reduce institutional resistance by demonstrating effectiveness before requiring substantial resource commitments. The assessment pack particularly addresses concerns about academic rigor by providing validated measurement tools that satisfy institutional accountability requirements.

9.4 Methodological strengths and innovations

The participatory design methodology employed in this study represents a methodological innovation in educational toolkit development. By including diverse stakeholder perspectives throughout the development process, we ensured that the final product addresses real-world needs rather than theoretical ideals. The three-phase iterative refinement process allowed for continuous improvement based on user feedback, resulting in a toolkit that demonstrates high usability and effectiveness. The multi-site European validation provides evidence of cross-cultural applicability, addressing concerns about generalizability that often limit educational intervention research. The inclusion of participants with diverse needs as co-designers rather than merely subjects represents an important advancement in inclusive design methodology, ensuring that accessibility solutions are developed with rather than for the communities they serve.

9.5 Future research directions

Building on the promising results of this study, several important avenues for future research would extend the TEGA framework's impact and address current limitations. Future research should examine the toolkit's effectiveness across diverse global contexts beyond Europe, particularly in educational systems with different structural approaches and cultural attitudes toward gaming. Comparative studies in Asian, African, and Latin American contexts could identify cultural adaptation

requirements and potentially reveal universal versus culturally-specific aspects of inclusive game design principles. While our 6-month follow-up provided valuable initial sustainability data, longitudinal studies tracking toolkit adoption, implementation patterns, and effectiveness over 2–3 year periods would yield crucial insights into long-term impact and the conditions that support sustained practice change. Such studies could identify institutional factors that promote persistent implementation and explore how practitioner expertise develops over extended periods. The promising results from analog implementations suggest significant potential for adapting the cybernetic-motivational framework to digital and hybrid game contexts. Research exploring how the TEGA principles translate to technology-mediated environments could bridge important gaps between analog and digital game-based learning while preserving the core inclusivity features that made the toolkit effective. Investigating how the TEGA framework might be adapted for different educational levels would significantly extend its impact. Studies examining age-appropriate modifications for primary, secondary, and professional education would help identify developmental considerations that influence inclusive game design across the lifespan, potentially creating a comprehensive developmental model of inclusive game-based learning. Research examining the organizational and institutional factors that support successful implementation would address the observed dependence on initial training and support. Studies could identify minimum viable training models, investigate peer mentoring approaches, and develop implementation toolkits that reduce institutional barriers to adoption, potentially increasing the framework's scalability across diverse resource environments. Future work should employ more rigorous sampling approaches with random assignment to address potential selection bias in the current study. Particularly valuable would be studies including educators with limited prior game-based learning experience and diverse attitudes toward educational gaming, providing stronger evidence of the toolkit's effectiveness across varied practitioner populations. These research directions would collectively strengthen the evidence base for the TEGA framework while extending its applicability across broader educational contexts, ultimately advancing the field's understanding of inclusive game-based learning principles. Building on the strong foundation established for analog game design, future research could productively explore how the TEGA framework might be adapted to digital gaming contexts. While this study intentionally focused on analog games due to their accessibility, cost-effectiveness, and flexibility in diverse educational settings (Boghian et al., 2019; Maratou et al., 2023), the cybernetic-motivational principles underlying the framework have potential applications across gaming modalities. Future work could investigate how these principles translate to digital environments, potentially creating an integrated framework that spans both analog and digital educational games while maintaining the core inclusivity features that made the toolkit effective. Such research would bridge important gaps between analog and digital game-based learning approaches, offering educators flexible implementation options based on available resources and specific learning contexts. The study's specific focus on higher education represents both a strength and a boundary for generalizability. By concentrating on higher education contexts, where inclusive game-based learning approaches are particularly underdeveloped (Dell et al., 2015;

Moriña, 2017), the research addresses a critical gap in the literature. This targeted approach allowed for deeper exploration of the unique challenges and requirements of diverse adult learners in tertiary settings. While findings may not directly transfer to other educational levels with different developmental considerations and institutional structures, this boundary establishes clear parameters for the toolkit's validated application. Future research could productively investigate the TEGA toolkit's effectiveness across other age groups, extending its impact beyond higher education while building on the strong foundation established in this adult learning context.

10 Conclusion

This study successfully developed and validated the TEGA Toolkit, representing the first comprehensive, evidence-based framework specifically designed for creating inclusive analog games in higher education. Through a rigorous three-phase participatory design process involving 35 stakeholders across four European universities, we demonstrated that educators can effectively implement inclusive game design principles when provided with appropriate theoretical frameworks and practical tools.

10.1 Key contributions

The research makes three primary contributions to the field: (1) development of a novel cybernetic-motivational design framework that integrates inclusivity as a core rather than peripheral consideration; (2) validation of 14 specific inclusivity measures across motor, cognitive, visual, and auditory domains; and (3) demonstration of significant improvements in educator capability and confidence in creating accessible educational games. These results substantially exceed typical educational intervention outcomes, suggesting that the toolkit addresses genuine practitioner needs while providing theoretically grounded solutions.

10.2 Broader implications

Beyond its immediate practical applications, this research contributes to broader conversations about educational equity and universal design for learning. By demonstrating that inclusive design principles can be systematically implemented without compromising educational effectiveness, the study challenges assumptions that accessibility accommodations necessarily reduce academic rigor or student engagement. The cybernetic-motivational framework offers a new paradigm for educational game design that prioritizes adaptability and responsiveness to diverse learner needs. This approach has implications beyond gaming contexts, suggesting how other educational technologies might be designed to automatically adjust to individual differences rather than requiring manual accommodations.

11 Limitations and future directions

While the study provides strong evidence for the toolkit's effectiveness within European higher education contexts, several limitations warrant consideration. The convenience sampling approach, focus on analog games, and relatively short follow-up period all limit generalizability. Additionally, the toolkit's effectiveness appears dependent on initial training and institutional support, suggesting implementation barriers that extend beyond individual educator capabilities.

Future research should examine long-term sustainability, cross-cultural applicability, and effectiveness across different educational levels. Investigation of digital game applications and cost-effectiveness compared to traditional accommodation approaches would further establish the toolkit's value proposition. Most importantly, research examining the relationship between inclusive game design and broader educational equity outcomes would demonstrate the toolkit's contribution to social justice goals in higher education.

12 Final recommendations

For educators interested in implementing inclusive game-based learning, this research provides both theoretical foundation and practical guidance. The TEGA Toolkit offers a systematic approach that moves beyond *ad hoc* accommodations to embedded inclusive design principles. However, successful implementation requires institutional commitment to providing training, resources, and ongoing support for educators.

For researchers, this study demonstrates the value of participatory design methodologies in educational technology development and suggests fruitful directions for continued investigation of adaptive, inclusive educational systems. The cybernetic-motivational framework provides a foundation for future research into responsive educational technologies that automatically adjust to diverse learner needs.

The TEGA Toolkit represents a significant step toward making higher education more inclusive and engaging for all students. By providing educators with evidence-based tools for creating accessible, effective educational games, this research contributes to the broader goal of educational equity while demonstrating the practical feasibility of universal design for learning principles in game-based contexts.

Data availability statement

The datasets presented in this article are not readily available due to copyright, and ethics approval restrictions. Requests to access the datasets should be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by London South Bank University Ethics Committee. The studies were conducted in accordance with the local legislation and institutional

requirements. The participants provided their written informed consent to participate in this study.

Author contributions

SR: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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